



**NAVSADRI
EDUCATION SOCIETY'S GROUP OF INSTITUTIONS, PUNE
DEPARTMENT OF AIML ENGINEERING
ACADEMIC YEAR: 2025-26**



Software Laboratory II

Artificial Neural Network

Lab Manual

TE AIML Semester -II

2025 – 26



**National Education Policy (NEP)-2020 Compliant
Curriculum**

**Third Year Engineering (2019 Pattern) in
Artificial Intelligence and Machine Learning
(With effect from Academic Year 2025-26)**

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ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to **Savitribai Phule Pune University, Pune**, for designing a comprehensive and industry-oriented curriculum in Artificial Intelligence and Machine Learning, which provided a strong academic foundation for this laboratory work.

I would like to express my sincere thanks to the **Head of the Department, Artificial Intelligence & Machine Learning Engineering, Prof. I. T. Mukherjee**, and the **Head of the Department, Computer Engineering, Prof. T. D. Kolhe**, for their valuable guidance, academic support, and encouragement throughout the completion of this laboratory manual.

I am highly grateful to our Respected **Principal, Dr. A. D. Lokhande**, for his valuable guidance, continuous motivation, and for granting permission and support to carry out this laboratory work successfully.

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I am thankful to all the faculty members and laboratory staff for their cooperation, technical assistance, and timely support during the laboratory sessions.

Finally, I would like to thank my colleagues and friends for their cooperation and support, which made the completion of this laboratory work a valuable and enriching learning experience.

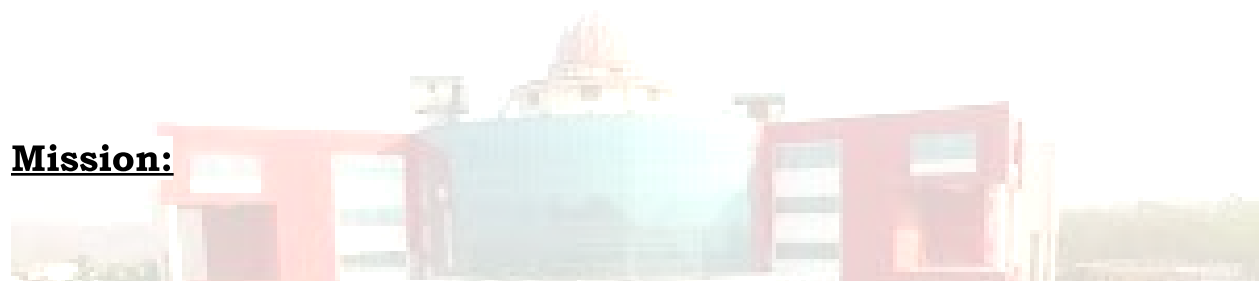


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ACADEMIC YEAR: 2025-26
Institute Vision & Mission



Vision:

The institute vision is to establish the centre for excellence in professional development and entrepreneurship development resulting into the enhancement of rural area. The institute also vision for developing professionals and citizens/citizenship to foster professional and rural development.



Mission:

NESGI offers skill based programs, graduate and post-graduate programmes in engineering and management to build competent manpower to suit the ever-changing requirements in industry and business by supporting students for continual development through excellence, technology based instructions and overall development of personality in all domains. The institute provides industry based education and practical training to the rural base.



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Departmental Vision & Mission

Vision:

To become a leading center of excellence in Artificial Intelligence and Machine Learning education, innovation, and research, cultivating ethically driven professionals who pioneer AI technologies to address global challenges and contribute to sustainable development, with a special focus on empowering rural and underserved communities.

Mission:

- The AIML Department is dedicated to providing cutting-edge education and fostering innovation in Artificial Intelligence and Machine Learning.
- Our mission is to equip students with the knowledge, skills, and ethical foundation to design intelligent systems that bridge societal gaps, improve quality of life, and create impactful solutions for regional, national, and global challenges.
- We strive to advance research, cultivate industry-academic collaborations, and promote the responsible and sustainable application of AI technologies to benefit humanity.



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Program Specific Outcomes (PSO)

A graduate of the Artificial Intelligence & Machine Learning Program will demonstrate-

PSO1	An ability to apply the theoretical concepts and practical knowledge of Artificial Intelligence & Machine Learning in analysis, design, development and management of information processing systems and applications in the interdisciplinary domain.
PSO2	An ability to analyze a problem, and identify and define the computing infrastructure and operations requirements appropriate to its solution. AI & ML graduates should be able to work on large-scale computing systems.
PSO3	An understanding of professional, business and business processes, ethical, legal, security and social issues and responsibilities.
PSO4	Practice communication and decision-making skills through the use of appropriate technology and be ready for professional responsibilities.





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PROGRAM OUTCOMES (POS)

Students are expected to know and be able to–

PO1	Engineering knowledge	An ability to apply knowledge of mathematics, computing, science, engineering and technology.
PO2	Problem analysis	An ability to define a problem and provide a systematic solution with the help of conducting experiments, analyzing the problem and interpreting the data.
PO3	Design / Development of Solutions	An ability to define a problem and provide a systematic solution with the help of conducting experiments, analyzing the problem and interpreting the data.
PO4	Conduct Investigations of Complex Problems	An ability to identify, formulates, and provides systematic solutions to complex engineering/Technology problems.
PO5	Modern Tool Usage	An ability to use the techniques, skills, and modern engineering technology tools, standard processes necessary for practice as a IT professional.
PO6	The Engineer and Society	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems with necessary constraints and assumptions.
PO7	Environment and Sustainability	An ability to analyze and provide solution for the local and global impact of information technology on individuals, organizations and society.
PO8	Ethics	An ability to understand professional, ethical, legal, security and social issues and responsibilities.
PO9	Individual and Team Work	An ability to function effectively as an individual or as a team member to accomplish a desired goal(s).
PO10	Communication Skills	An ability to engage in life-long learning and continuing professional development to cope up with fast changes in the technologies/tools with the help of electives, professional organizations and extra-curricular activities.
PO11	Project Management and Finance	An ability to communicate effectively in engineering community at large by means of effective presentations, report writing, paper publications, demonstrations.
PO12	Life-long Learning	An ability to understand engineering, management, financial aspects, performance, optimizations and time complexity necessary for professional practice.



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Course Objectives:

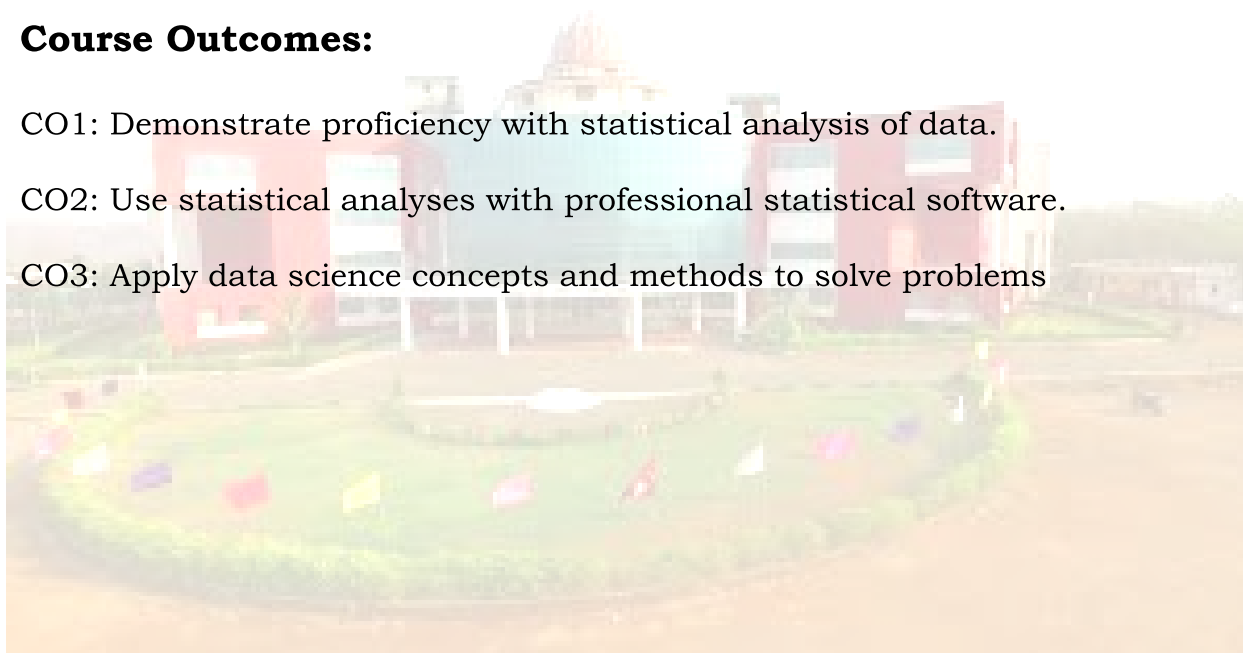
1. Students will demonstrate proficiency with statistical analysis of data.
2. Students will execute statistical analyses with professional statistical software.
3. Students will apply data science concepts and methods to solve problems.

Course Outcomes:

CO1: Demonstrate proficiency with statistical analysis of data.

CO2: Use statistical analyses with professional statistical software.

CO3: Apply data science concepts and methods to solve problems



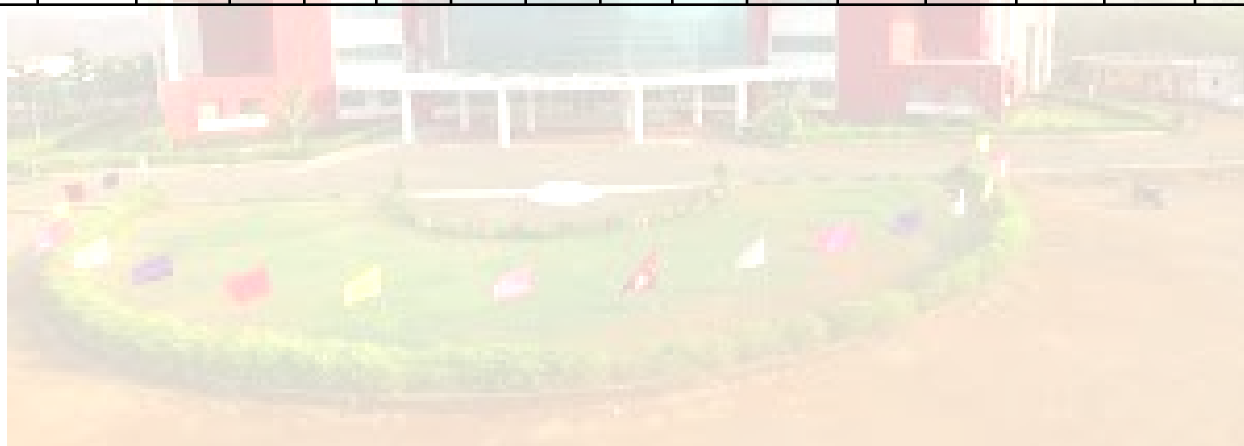


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CO-PO-PSO Mapping for ANN Practical

CO \ PO / PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	1	2	1	2	1	1	1	1	1	1	3	2	1	1
CO2	2	2	2	1	3	1	1	1	2	2	2	2	3	3	1	2
CO3	3	3	3	3	3	2	2	2	2	2	2	3	3	3	2	3





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Third Year of Artificial Intelligence and Machine Learning (2019 Course)
Course Code: 318556 Software Laboratory II
Course Name: Artificial Neural Network Lab
Experiment No. 1

Title

Implementation and graphical demonstration of commonly used activation functions in Artificial Neural Networks using Python.

Aim

To study, implement, and visualize different activation functions used in Artificial Neural Networks and analyze their behavior using graphical representation.

Objectives

After completion of this experiment, students will be able to:

1. Understand the role of activation functions in neural networks.
2. Implement different activation functions using Python.
3. Analyze non-linear transformations through graphical visualization.
4. Relate activation functions to learning capability of neural networks.

Course Outcomes (COs) Addressed

CO Code	Description
CO1	Demonstrate proficiency with statistical analysis of data
CO2	Use statistical analyses with professional statistical software
CO3	Apply data science concepts and methods to solve problems

CO-PO-PSO Mapping Table

CO \ PO / PSO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	✓	✓				✓	
CO2	✓	✓	✓		✓	✓	✓
CO3	✓	✓	✓	✓	✓	✓	✓

✓ – Strong Mapping

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Storage: Minimum 20 GB free space
- Display: Standard Monitor
- Input Devices: Keyboard and Mouse

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Course Name: Artificial Neural Network Lab
Software Requirements

- Operating System: Windows 10 / Linux / macOS
 - Programming Language: Python 3.8 or above
 - IDE/Editor: Anaconda Navigator / Spyder / VS Code / PyCharm
- Libraries:
- NumPy
 - Matplotlib

Theory

Artificial Neural Networks (ANNs) are computational models inspired by the biological neural system of the human brain. An ANN consists of interconnected processing units called neurons, which process information using weighted connections. Each neuron computes a weighted sum of inputs and passes it through an **activation function** to generate an output.

Role of Activation Functions

Activation functions introduce **non-linearity** into neural networks. Without activation functions, a neural network would behave like a simple linear model regardless of the number of layers. Non-linearity allows neural networks to learn complex patterns, relationships, and decision boundaries.

Mathematically, the output of a neuron is given by:

$$y = f \left(\sum_{i=1}^n w_i x_i + b \right)$$

where:

- x_i are inputs
- w_i are weights
- b is bias
- $f(.)$ is the activation function

Types of Activation Functions Implemented

1. Step Function

- Output is binary (0 or 1).
- Used in early perceptron models.
- Not differentiable → unsuitable for backpropagation.

$$f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

2. Sigmoid Function

- Produces output in range (0,1).
- Smooth and differentiable.
- Commonly used in binary classification.

$$f(x) = \frac{1}{1 + e^{-x}}$$

3. Hyperbolic Tangent (Tanh)

- Output range (-1,1).
- Zero-centered → better convergence than sigmoid.

$$f(x) = \tanh(x)$$

4. Rectified Linear Unit (ReLU)

- Most widely used activation function.
- Efficient computation.
- Solves vanishing gradient problem.

$$f(x) = \max(0, x)$$

Importance in AI & ML

Activation functions enable neural networks to:

- Perform classification and regression tasks
- Model complex real-world data
- Improve learning efficiency and accuracy

Algorithm

1. Import required Python libraries (NumPy and Matplotlib).
2. Generate a range of input values.
3. Define mathematical expressions for activation functions.
4. Compute output values for each function.
5. Plot graphs for all activation functions.
6. Analyze behavior of each activation function.

Case Study

Consider a medical diagnosis system for disease prediction.

- Sigmoid activation is used to predict probability of disease (Yes/No).
- ReLU is used in hidden layers to speed up training.
- Tanh helps in learning symmetric data patterns.

Thus, selection of appropriate activation functions directly impacts accuracy and performance of AI systems.

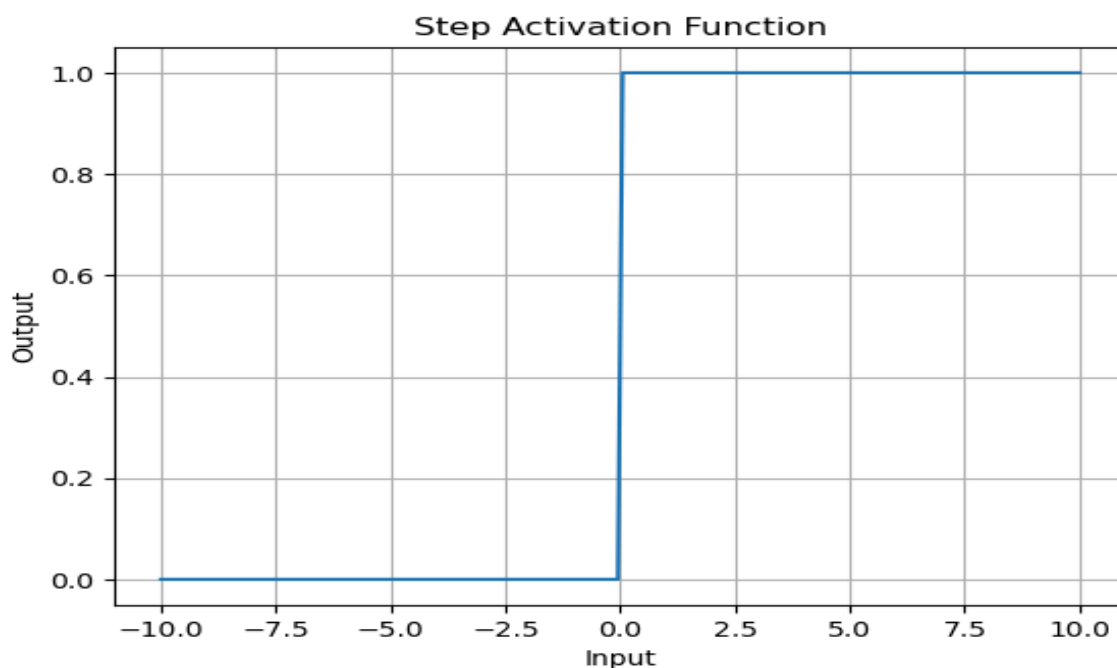
Program Python Code: Separate Graph for Each Activation Function

Cell 1: Import Libraries & Input Range

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(-10, 10, 200)
```

1.Step Activation Function (Separate Graph)

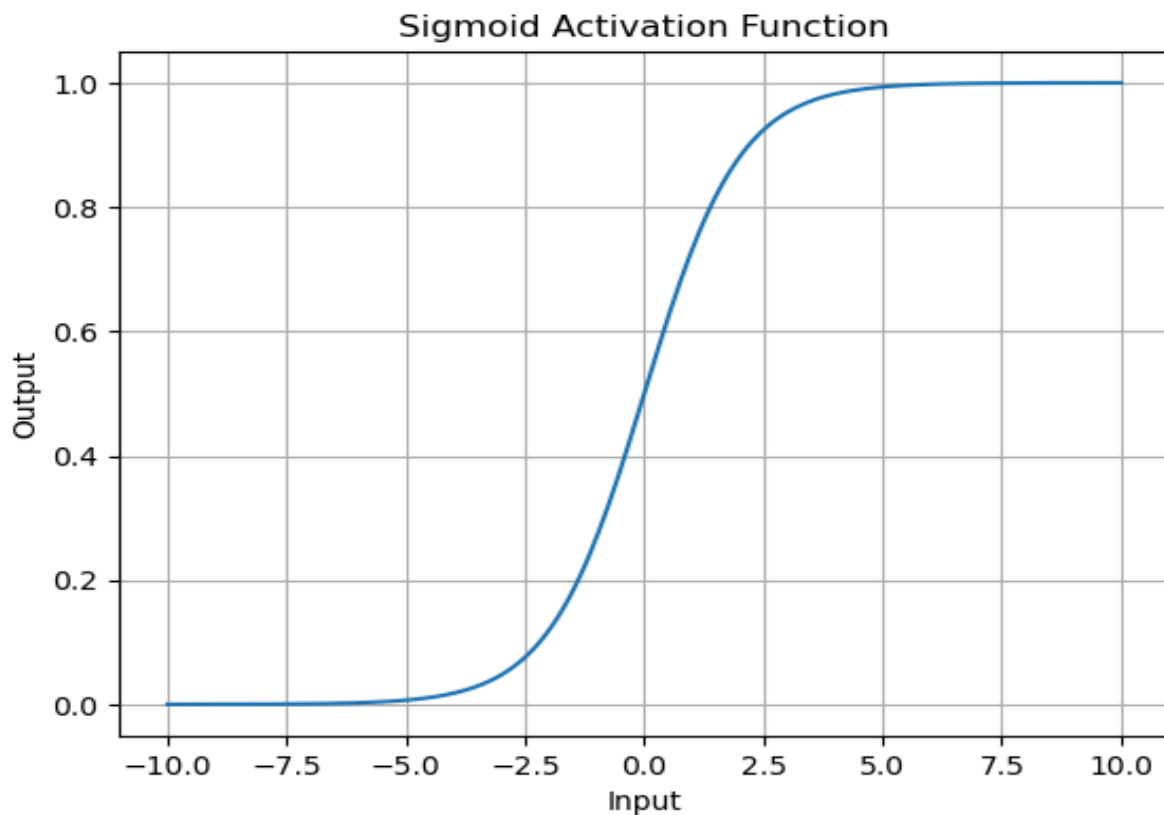
```
step = np.where(x >= 0, 1, 0)
plt.figure()
plt.plot(x, step)
plt.title("Step Activation Function")
plt.xlabel("Input")
plt.ylabel("Output")
plt.grid()
plt.show()
```



“This function only gives YES or NO. That’s why early neural networks were limited.”

2. Sigmoid Activation Function (Separate Graph)

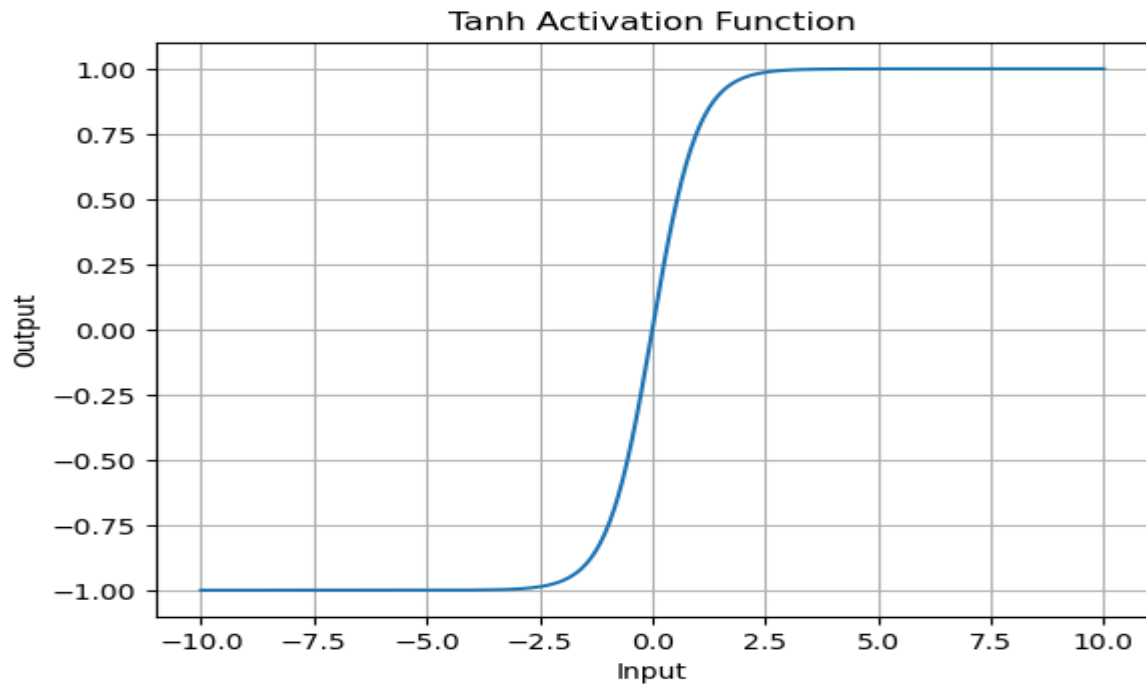
```
sigmoid = 1 / (1 + np.exp(-x))  
plt.figure()  
plt.plot(x, sigmoid)  
plt.title("Sigmoid Activation Function")  
plt.xlabel("Input")  
plt.ylabel("Output")  
plt.grid()  
plt.show()
```



“Notice how learning slows down at extreme values — this is called saturation.”

3. Tanh Activation Function (Separate Graph)

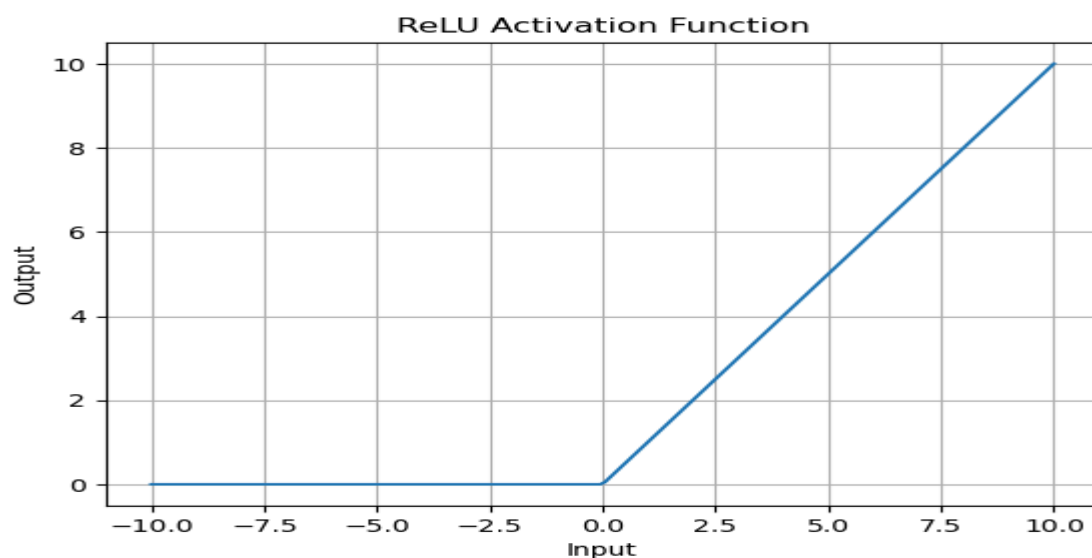
```
tanh = np.tanh(x)  
plt.figure()  
plt.plot(x, tanh)  
plt.title("Tanh Activation Function")  
plt.xlabel("Input")  
plt.ylabel("Output")  
plt.grid()  
plt.show()
```



“Tanh is zero-centered, so it learns faster than sigmoid.”

4. ReLU Activation Function (Separate Graph)

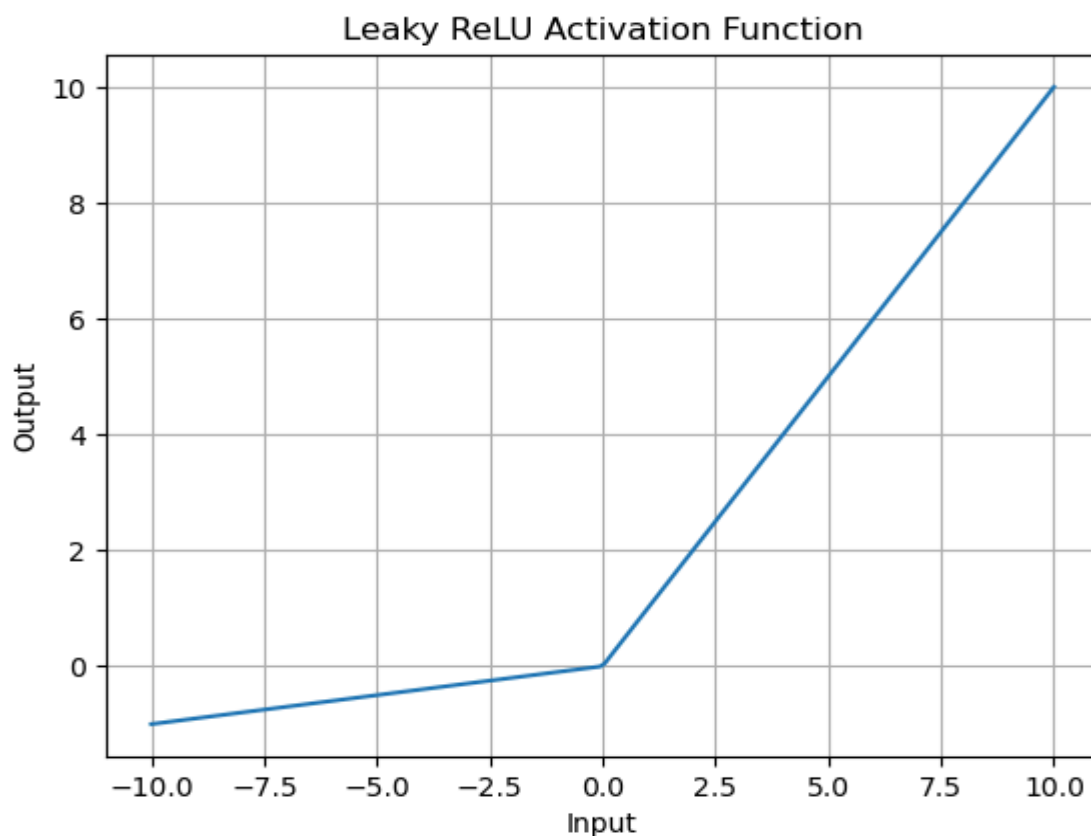
```
relu = np.maximum(0, x)
plt.figure()
plt.plot(x, relu)
plt.title("ReLU Activation Function")
plt.xlabel("Input")
plt.ylabel("Output")
plt.grid()
plt.show()
```



“This single line made deep learning practical.”

5 Leaky ReLU Activation Function (Separate Graph – EXTRA)

```
leaky_relu = np.where(x > 0, x, 0.1*x)
plt.figure()
plt.plot(x, leaky_relu)
plt.title("Leaky ReLU Activation Function")
plt.xlabel("Input")
plt.ylabel("Output")
plt.grid()
plt.show()
```



Advanced concept

“Leaky ReLU solves the dead neuron problem of ReLU.”

Observation

- Step function produces binary output.
- Sigmoid and tanh provide smooth curves.
- ReLU outputs zero for negative inputs and linear output for positive inputs.

- ReLU shows computational efficiency.

Output

Graphical plots showing:

- Step function
- Sigmoid function
- Tanh function
- ReLU function

Viva Questions

1. What is the purpose of activation functions?
2. Why is ReLU preferred over sigmoid?
3. What is vanishing gradient problem?
4. Which activation function is used in output layer for classification?
5. Can neural networks work without activation functions?

Conclusion

In this experiment, various activation functions used in Artificial Neural Networks were successfully implemented and analyzed. The graphical visualization helped in understanding their behavior and importance in learning non-linear data patterns. Activation functions play a crucial role in improving the learning capability and performance of neural networks used in AI and ML applications.

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Savitribai Phule Pune University
Third Year of Artificial Intelligence and Machine Learning (2019 Course)
Course Code: 318556 Software Laboratory II
Course Name: Artificial Neural Network Lab
Experiment No. 2

Title

Implementation of Back Propagation Neural Network for XOR Function with Binary Input and Output using Python.

Aim

To design and implement a back propagation neural network to solve the XOR (Exclusive-OR) problem using binary inputs and outputs.

Objectives

After completion of this experiment, students will be able to:

1. Understand the concept of back propagation learning.
2. Analyze why XOR problem is non-linearly separable.
3. Implement a multi-layer neural network using Python.
4. Validate learning capability of neural networks using XOR logic.

Course Outcomes (COs) Addressed

CO Code	Description
CO1	Demonstrate proficiency with statistical analysis of data
CO2	Use statistical analyses with professional statistical software
CO3	Apply data science concepts and methods to solve problems

CO-PO-PSO Mapping Table

CO \ PO / PSO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	✓	✓				✓	
CO2	✓	✓	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓	✓	✓

✓ – Strong Mapping

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: Minimum 20 GB free space
- Input Devices: Keyboard, Mouse

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Course Name: Artificial Neural Network Lab

- Output Device: Monitor

Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.8 or above
- IDE: Anaconda / Spyder / VS Code
- Python Libraries:
 - NumPy
 - Scikit-learn

Theory

Artificial Neural Networks learn by adjusting weights to minimize the error between predicted output and actual output. **Back propagation** is the most widely used supervised learning algorithm for training multi-layer neural networks.

XOR Problem

The XOR logic function produces output **1 only when inputs are different**, otherwise output is **0**.

Input X1	Input X2	XOR Output
0	0	0
0	1	1
1	0	1
1	1	0

The XOR problem **cannot be solved using a single-layer perceptron** because the data is **not linearly separable**. This limitation led to the development of **multi-layer perceptrons (MLPs)** and the **back propagation algorithm**.

Back Propagation Algorithm

Back propagation works in two main phases:

1. Forward Propagation

- Inputs are passed through the network.
- Weighted sums are calculated.
- Activation functions generate outputs.

2. Backward Propagation

- Error is calculated at output layer.
- Error is propagated backward.
- Weights are updated using gradient descent.

The weight update rule is:

$$w_{new} = w_{old} - \eta \frac{\partial E}{\partial w}$$

Where:

- η = learning rate
- E = error function

Importance of XOR Experiment

- Demonstrates necessity of hidden layers.
- Validates learning capability of neural networks.
- Forms the foundation for deep learning models.

Algorithm

1. Define XOR input patterns and target outputs.
2. Initialize a multi-layer perceptron network.
3. Specify hidden layer size and activation function.
4. Train the network using back propagation.
5. Test the trained model using XOR inputs.
6. Display predicted outputs.

Case Study

Consider a **binary decision system** such as fraud detection:

- Single parameter checks may fail.
- Combination of parameters using hidden layers enables correct classification.

Similarly, XOR demonstrates how neural networks learn complex decision boundaries.

Program (Python Code)

```
import numpy as np
from sklearn.neural_network import MLPClassifier
# XOR input and output
X = np.array([[0,0], [0,1], [1,0], [1,1]])
y = np.array([0, 1, 1, 0])
```

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```
# Create MLP model
model = MLPClassifier(
    hidden_layer_sizes=(2,),
    activation='logistic',
    solver='sgd',
    learning_rate_init=0.5,
    max_iter=5000
)
# Train the model
model.fit(X, y)
# Predict output
predictions = model.predict(X)
print("Input XOR Output")
for i in range(len(X)):
    print(X[i], "->", predictions[i])
```

Observation

- Single-layer perceptron fails for XOR.
- Multi-layer perceptron successfully learns XOR.
- Hidden layer enables non-linear decision boundary.

Output

Input XOR Output

```
[0 0] -> 0
[0 1] -> 1
[1 0] -> 1
[1 1] -> 0
```

Viva Questions

1. Why XOR cannot be solved using single-layer perceptron?
2. What is back propagation?
3. What is the role of hidden layer?
4. Which activation function is used here?
5. What happens if learning rate is very high?

Conclusion

In this experiment, a back propagation neural network was successfully implemented to solve the XOR problem. The results clearly demonstrate the necessity of hidden layers and non-linear activation functions in neural networks. This experiment validates the learning capability of multi-layer perceptrons and forms a foundation for advanced neural network architectures.

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Course Code: 318556 Software Laboratory II
Course Name: Artificial Neural Network Lab
Experiment No. 3

Title

Implementation of a Feed Forward Neural Network using Back Propagation Learning Algorithm in Python.

Aim

To design and implement a feed forward neural network trained using back propagation algorithm for classification of real-world data.

Objectives

After completion of this experiment, students will be able to:

1. Understand the architecture of feed forward neural networks.
2. Implement back propagation learning for multi-class classification.
3. Analyze performance of neural networks on real datasets.
4. Apply neural network concepts to practical AI & ML problems.

Course Outcomes (COs) Addressed

CO Code	Description
CO1	Demonstrate proficiency with statistical analysis of data
CO2	Use statistical analyses with professional statistical software
CO3	Apply data science concepts and methods to solve problems

CO-PO-PSO Mapping Table

CO \ PO / PSO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	✓	✓				✓	
CO2	✓	✓	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓	✓	✓

✓ – Strong Mapping

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Storage: Minimum 20 GB free space
- Input Devices: Keyboard, Mouse
- Output Device: Monitor

Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.8 or above

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Course Code: 318556 Software Laboratory II
Course Name: Artificial Neural Network Lab

- IDE: Anaconda / Spyder / VS Code
- Python Libraries:
 - NumPy
 - Scikit-learn

Theory

A **Feed Forward Neural Network (FFNN)** is one of the simplest and most widely used neural network architectures. In this network, information flows in **one direction only**, from the input layer through hidden layers to the output layer. There are no cycles or feedback connections.

Architecture of Feed Forward Neural Network

A typical FFNN consists of:

1. **Input Layer** – Receives raw data features.
2. **Hidden Layer(s)** – Performs intermediate computations.
3. **Output Layer** – Produces final prediction.

Each neuron computes a weighted sum of its inputs, adds a bias, and applies an activation function.

Back Propagation Learning

Back propagation is a supervised learning algorithm used to minimize error by updating network weights. It works based on **gradient descent optimization**.

The learning process consists of:

- Forward pass: Compute network output.
- Error computation: Difference between actual and predicted output.
- Backward pass: Adjust weights to minimize error.

Mathematical Representation

Error function:

$$E = \frac{1}{2} \sum (t - y)^2$$

Weight update:

$$\Delta w = -\eta \frac{\partial E}{\partial w}$$

Where:

- t = target output
- y = predicted output
- η = learning rate

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Course Name: Artificial Neural Network Lab
Application in AI & ML

Feed forward neural networks are used in:

- Pattern recognition
- Image classification
- Speech recognition
- Medical diagnosis
- Financial forecasting

Algorithm

1. Load dataset.
2. Separate input features and target labels.
3. Split data into training and testing sets.
4. Create a feed forward neural network model.
5. Train model using back propagation.
6. Evaluate model accuracy.

Case Study

Consider an **automated flower classification system**:

- Input features: Petal length, petal width, sepal length, sepal width.
- Output: Flower species.
- FFNN learns complex relationships between features.

This experiment demonstrates how neural networks can be used in real-world classification tasks.

Program (Python Code)

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier

# Load dataset
data = load_iris()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Feed Forward Neural Network
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation='relu',
```

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Course Name: Artificial Neural Network Lab

```
solver='adam',  
max_iter=2000  
)  
  
# Train model  
model.fit(X_train, y_train)  
  
# Evaluate accuracy  
accuracy = model.score(X_test, y_test)  
print("Model Accuracy:", accuracy)
```

Observation

- Feed forward neural network successfully classifies multi-class data.
- Back propagation efficiently reduces classification error.
- ReLU activation improves convergence speed.

Output

Model Accuracy: 0.93 (approx.)

Viva Questions

1. What is a feed forward neural network?
2. Difference between feed forward and feedback networks?
3. Why is back propagation required?
4. Role of hidden layers?
5. What is gradient descent?

Conclusion

In this experiment, a feed forward neural network trained using back propagation algorithm was successfully implemented. The network demonstrated high classification accuracy on real-world data, proving the effectiveness of feed forward architectures in solving complex AI & ML problems.

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Third Year of Artificial Intelligence and Machine Learning (2019 Course)
Course Code: 318556 Software Laboratory II
Course Name: Artificial Neural Network Lab
Experiment No. 4

Title

Implementation and demonstration of Adaptive Resonance Theory (ART) Neural Network for clustering using Python.

Aim

To study and implement Adaptive Resonance Theory (ART) neural network for unsupervised learning and clustering of input patterns.

Objectives

After completion of this experiment, students will be able to:

1. Understand the concept of unsupervised learning.
2. Explain the working principle of ART neural networks.
3. Implement ART-based clustering using Python.
4. Analyze stability-plasticity trade-off in neural networks.

Course Outcomes (COs) Addressed

CO Code	Description
CO1	Demonstrate proficiency with statistical analysis of data
CO2	Use statistical analyses with professional statistical software
CO3	Apply data science concepts and methods to solve problems

CO-PO-PSO Mapping Table

CO \ PO / PSO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	✓	✓				✓	
CO2	✓	✓	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓	✓	✓

✓ – Strong Mapping

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: Minimum 20 GB free space
- Input Devices: Keyboard, Mouse
- Output Device: Monitor

Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.8 or above

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- IDE: Anaconda / Spyder / VS Code
- Python Libraries:
 - NumPy

Theory

Adaptive Resonance Theory (ART) is a neural network model developed to address the **stability-plasticity dilemma** in learning systems. The dilemma refers to the challenge of learning new information (plasticity) without forgetting previously learned information (stability).

ART networks are **unsupervised learning models** primarily used for **clustering and pattern recognition**. Unlike supervised learning, ART does not require labeled data and automatically determines the number of clusters based on similarity between input patterns.

Architecture of ART Neural Network

An ART network consists of:

1. **Comparison Layer (F1)** – Receives input patterns.
2. **Recognition Layer (F2)** – Represents learned clusters.
3. **Vigilance Parameter (ρ)** – Controls similarity threshold.

The vigilance parameter determines whether an input pattern matches an existing cluster or forms a new cluster.

Working Principle

1. Input pattern is presented to the F1 layer.
2. Similarity is computed with existing clusters.
3. If similarity $\geq$ vigilance threshold, resonance occurs.
4. Weights are updated to include new pattern.
5. If no match is found, a new cluster is created.

Advantages of ART Networks

- Stable learning
- Dynamic cluster formation
- No catastrophic forgetting
- Suitable for real-time systems

Applications

- Pattern recognition
- Image segmentation
- Document clustering

- Market segmentation

Algorithm

1. Initialize input patterns.
2. Initialize empty cluster list.
3. Set vigilance parameter.
4. For each input pattern:
 - Compare with existing clusters.
 - Check similarity measure.
 - Assign to cluster or create new cluster.
5. Display number of clusters formed.

Case Study

Consider **customer behavior analysis** in e-commerce:

- Customer purchase patterns are unlabeled.
- ART clusters customers dynamically.
- New customer behavior does not disturb existing clusters.

This makes ART suitable for evolving real-world data.

Program (Python Code)

```
import numpy as np

# Input patterns
patterns = np.array([
    [1, 0, 1, 0],
    [1, 1, 0, 0],
    [0, 0, 1, 1],
    [0, 1, 0, 1]
])

vigilance = 0.5
clusters = []

for p in patterns:
    matched = False
    for c in clusters:
        similarity = np.sum(p & c) / np.sum(p)
        if similarity >= vigilance:
            matched = True
            break
    if not matched:
        clusters.append(p)
```

```
print("Number of clusters formed:", len(clusters))
```

Observation

- ART network forms clusters dynamically.
- Higher vigilance value results in more clusters
- ART maintains stability while learning new patterns.

Output

Number of clusters formed: 2 (may vary based on vigilance)

Viva Questions

1. What is stability-plasticity dilemma?
2. What is the role of vigilance parameter?
3. Difference between ART and k-means clustering?
4. Is ART supervised or unsupervised?
5. Where is ART used in real-world applications?

Conclusion

In this experiment, an Adaptive Resonance Theory neural network was successfully implemented to demonstrate unsupervised learning and clustering. The experiment highlights the importance of stability and adaptability in neural networks and shows how ART efficiently handles evolving input data without forgetting learned patterns.

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Experiment No. 5

Title

Implementation of Perceptron Learning Law and visualization of its Decision Region using Python.

Aim

To implement the perceptron learning algorithm and demonstrate its decision boundary using graphical representation.

Objectives

After completion of this experiment, students will be able to:

1. Understand the concept of single-layer perceptron.
2. Implement perceptron learning law.
3. Analyze linear separability of data.
4. Visualize decision regions using graphical plots.
5. Relate theoretical learning rule with practical classification.

Course Outcomes (COs) Addressed

CO Code	Description
CO1	Demonstrate proficiency with statistical analysis of data
CO2	Use statistical analyses with professional statistical software
CO3	Apply data science concepts and methods to solve problems

CO-PO-PSO Mapping Table

CO \ PO / PSO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	✓	✓				✓	
CO2	✓	✓	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓	✓	✓

✓ – Strong Mapping

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Storage: Minimum 20 GB free space
- Input Devices: Keyboard, Mouse
- Output Device: Monitor

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Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.8 or above
- IDE: Anaconda / Spyder / VS Code
- Python Libraries:
 - NumPy
 - Matplotlib
 - Scikit-learn

Theory

The **Perceptron** is the simplest form of an artificial neural network, introduced by **Frank Rosenblatt**. It is a **single-layer, supervised learning model** used for binary classification of linearly separable data.

A perceptron consists of:

- Input layer
- Weight vector
- Bias
- Activation function (Step function)

The perceptron computes a weighted sum of inputs and applies an activation function to produce output.

Mathematical Model

$$y = f \left(\sum_{i=1}^n w_i x_i + b \right)$$

Where:

- x_i = input features
- w_i = weights
- b = bias
- f = activation function

Perceptron Learning Law

The perceptron updates its weights whenever it misclassifies a data point.

$$w_{new} = w_{old} + \eta(t - y)x$$
$$b_{new} = b_{old} + \eta(t - y)$$

Where:

- η = learning rate
- t = target output
- y = predicted output

Decision Region

A decision region divides the input space into areas corresponding to different output classes.

For a perceptron, this region is a **straight line (hyperplane)**.

Limitations

- Works only for linearly separable data.
- Cannot solve XOR problem.
- Requires careful learning rate selection.

Applications

- Binary classification
- Spam detection
- Simple pattern recognition
- Threshold-based decision systems

Algorithm

1. Define input dataset and target labels.
2. Initialize perceptron model.
3. Train the model using perceptron learning law.
4. Generate mesh grid for visualization.
5. Predict outputs over grid points.
6. Plot decision region and training data.

Case Study

Consider **email spam detection**:

- Inputs: frequency of keywords.
- Output: spam or not spam.
- If data is linearly separable, perceptron efficiently classifies emails.

This experiment demonstrates the practical usefulness of perceptron learning.

Program (Python Code)

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import Perceptron

# Training data
X = np.array([[1,1], [2,2], [3,3], [1,3], [2,4], [3,5]])
y = np.array([0, 0, 0, 1, 1, 1])

# Train perceptron
model = Perceptron(max_iter=1000, eta0=0.1)
```

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```
model.fit(X, y)
```

```
# Create decision region
xx, yy = np.meshgrid(
    np.linspace(0, 4, 200),
    np.linspace(0, 6, 200)
)

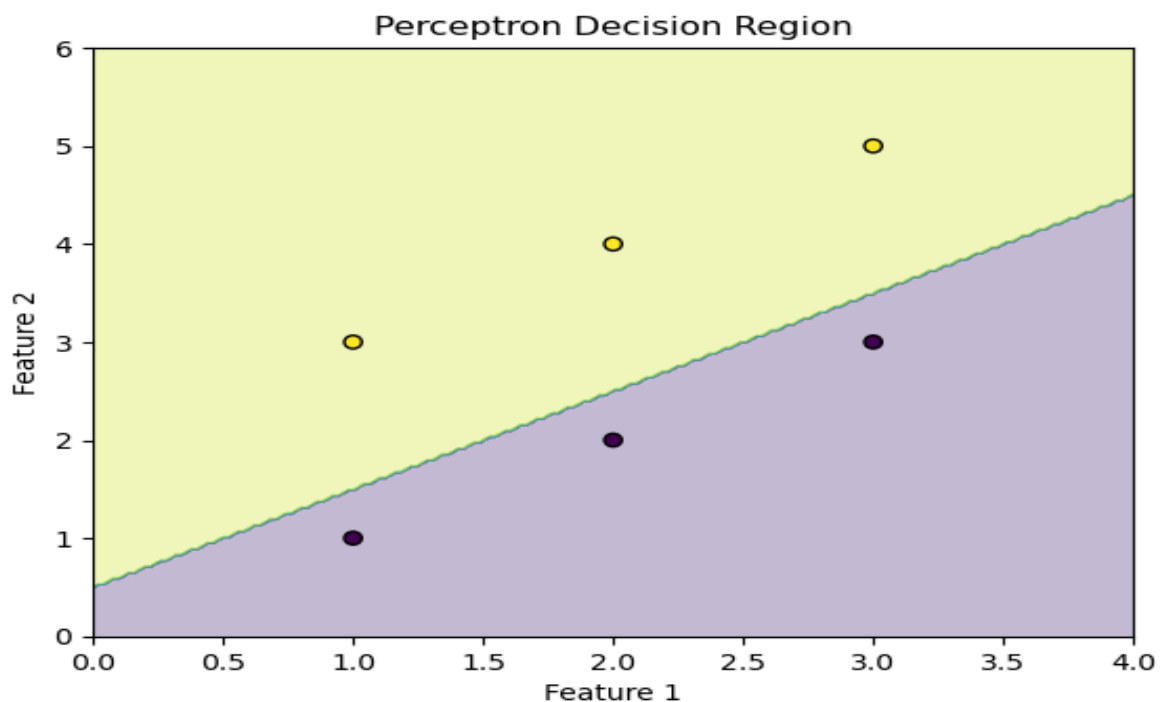
Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

# Plot
plt.contourf(xx, yy, Z, alpha=0.3)
plt.scatter(X[:,0], X[:,1], c=y, edgecolor='k')
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("Perceptron Decision Region")
plt.show()
```

Observation

- Perceptron correctly classifies linearly separable data.
- The decision boundary is a straight line.
- Graphical visualization clearly shows decision regions.

Output



Graph showing:

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- a. Training data points
- b. Linear decision boundary
- c. Separate decision regions for two classes

Viva Questions

- 1. What is a perceptron?
- 2. Explain perceptron learning law.
- 3. Why perceptron fails for XOR?
- 4. What is decision region?
- 5. Difference between perceptron and MLP?

Conclusion

In this experiment, the perceptron learning algorithm was successfully implemented and its decision region was visualized graphically. The experiment highlights the capability and limitations of single-layer neural networks and forms the foundation for understanding advanced neural network architectures.